



Final Documentation Report

On Project “House Booking System”

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Section: H1

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Course Name: Software Development Capstone Project

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1. Introduction

The House Booking System is a console-based application developed in the C programming language for managing house rental and booking operations. The system provides separate functionalities for administrators and customers. Administrators can manage house records and monitor bookings, while customers can search available houses, book houses, and manage their reservations.

The project was developed as an academic capstone project during the 2nd year of undergraduate studies to demonstrate practical implementation of programming concepts such as file handling, structures, functions, and menu-driven application development.

2. Problem Statement

Managing house rental information manually can be time-consuming and inefficient. Customers often face difficulties in searching for available houses and maintaining booking records, while administrators struggle to organize house information and monitor bookings efficiently.

The absence of a centralized system may lead to:

- Difficulty in managing house records
- Errors in booking management
- Time-consuming manual operations
- Poor organization of customer booking information

Therefore, a digital system is needed to simplify house management and booking processes.

3. Objectives

The main objectives of the project are:

- To develop a house booking management system using C
- To simplify the process of searching and booking houses
- To allow administrators to manage house information efficiently
- To maintain booking records in an organized manner
- To implement file handling for persistent data storage
- To improve understanding of structured programming concepts

4. Stakeholders

The main stakeholders of the system are:

1. Administrator

The administrator manages the entire system, including:

- Adding new houses
- Updating house information
- Viewing bookings
- Deleting house records

2. Customers

Customers use the system to:

- View available houses
- Search houses
- Book houses
- View and cancel bookings

5. Scope of the Project

In-Scope Features: The scope of the House Booking System includes-

- Managing house rental information
- Searching available houses
- Booking and canceling reservations
- Maintaining booking records using file handling
- Providing separate modules for admin and customers

Out-of-Scope Features: The project is limited to a console-based environment and does not include-

- Online payment integration
- Graphical user interface (GUI)
- Database management systems

- Network-based multi-user access

6. Tech Stack

Component	Technology
Programming Language	C
Application Type	Console-Based Application
Storage System	File Handling
Development Concepts	Structures, Functions, Loops, Conditional Statements

7. Functional Requirements

Admin Functional Requirements

1. The system shall allow the admin to add new houses.
2. The system shall allow the admin to view all houses.
3. The system shall allow the admin to view all bookings.
4. The system shall allow the admin to delete house records.
5. The system shall allow the admin to modify house details.

Customer Functional Requirements

1. The system shall allow customers to view available houses.
2. The system shall allow customers to search houses.
3. The system shall allow customers to book a house.
4. The system shall allow customers to view booking records.
5. The system shall allow customers to cancel bookings.

8. Non-Functional Requirements

Usability

The system should provide a simple and user-friendly menu-driven interface.

Reliability

The system should store and retrieve booking information accurately using file handling.

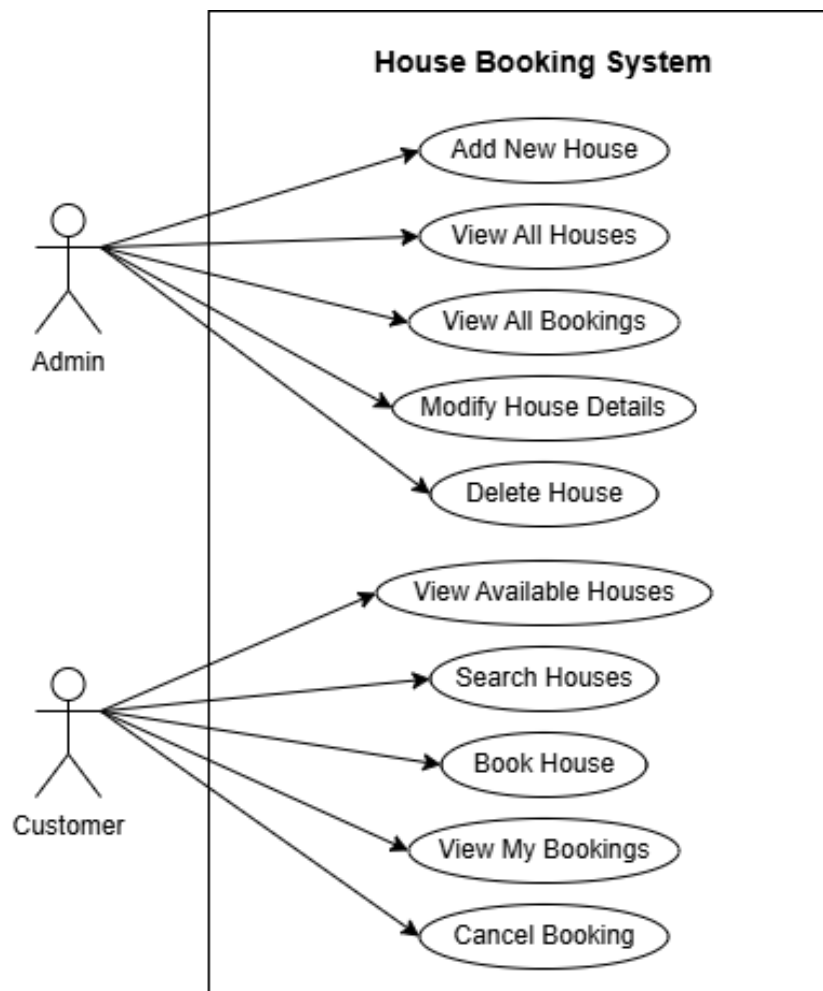
Maintainability

The source code should be organized and easy to modify for future improvements.

Portability

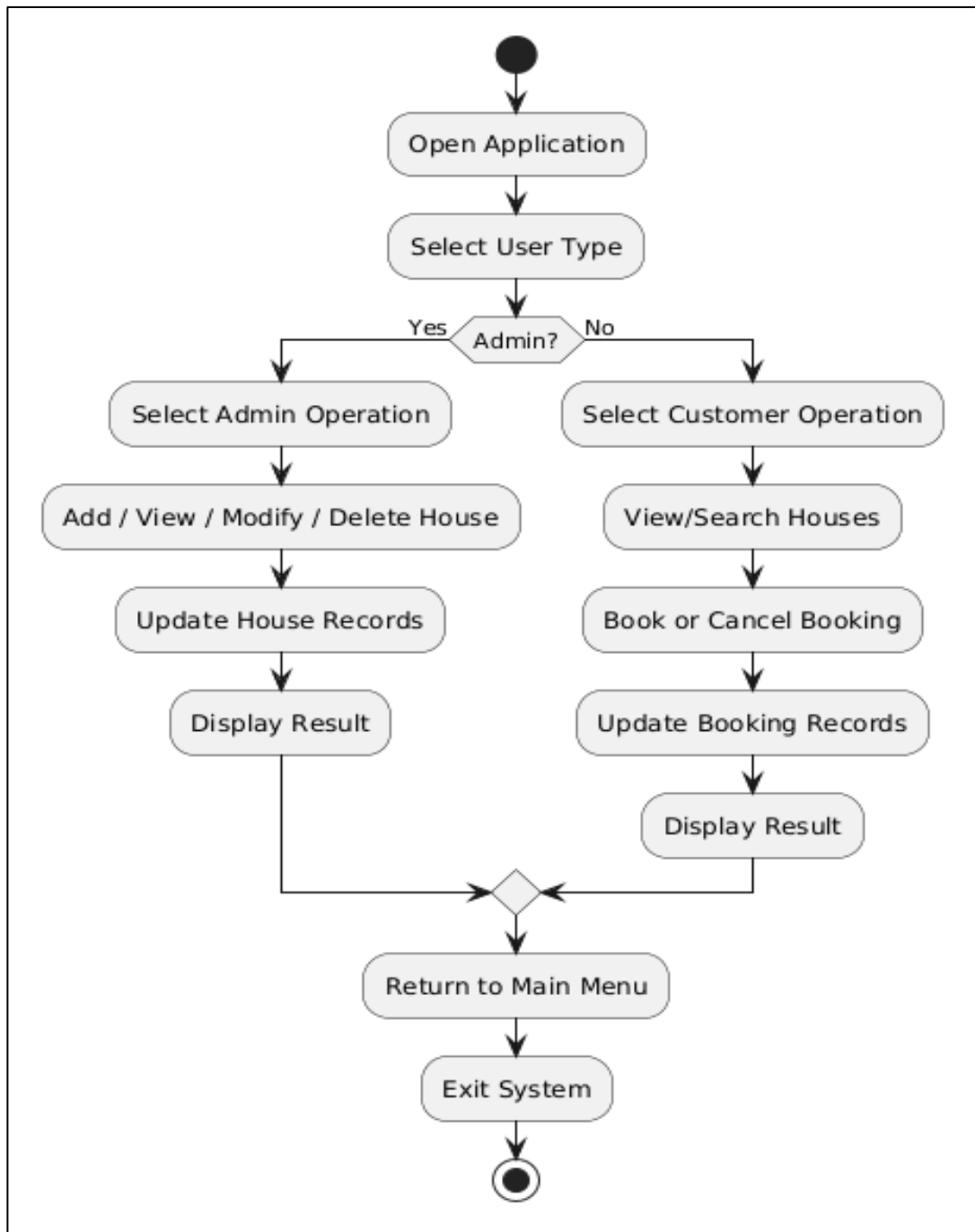
The application should run on systems supporting C compilers such as GCC or CodeBlocks.

9. Use Case Diagram

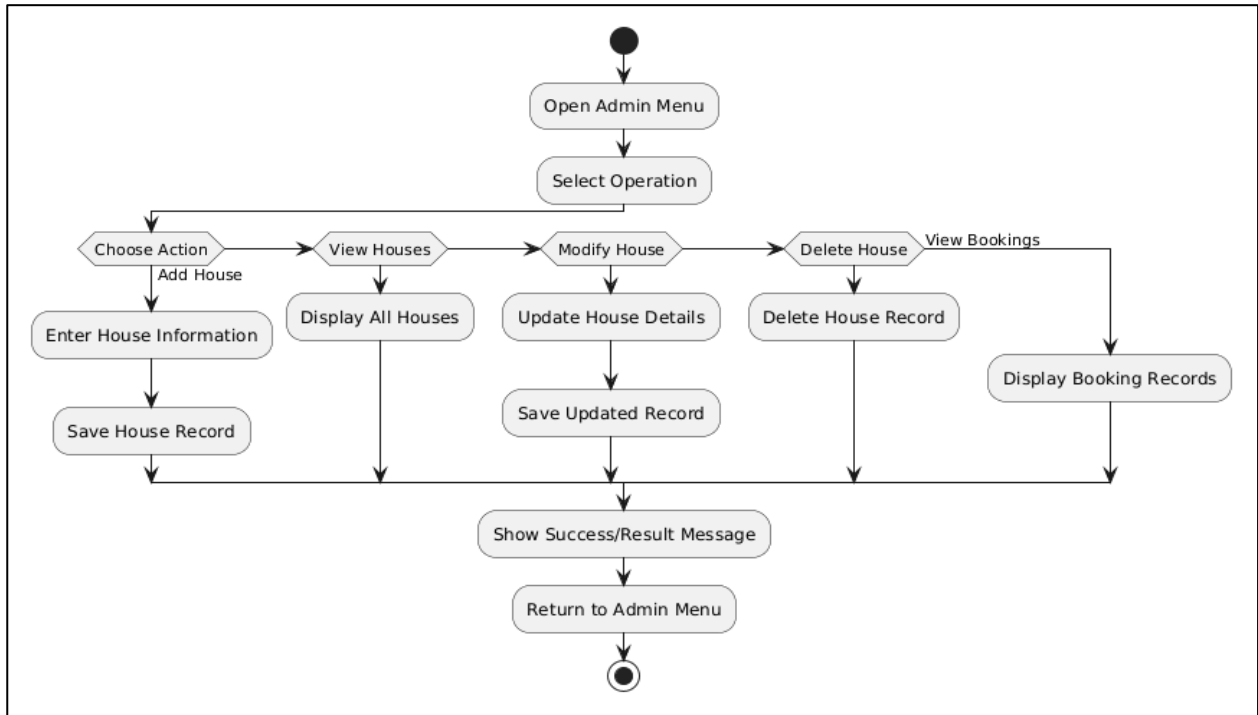


10. Activity Diagram

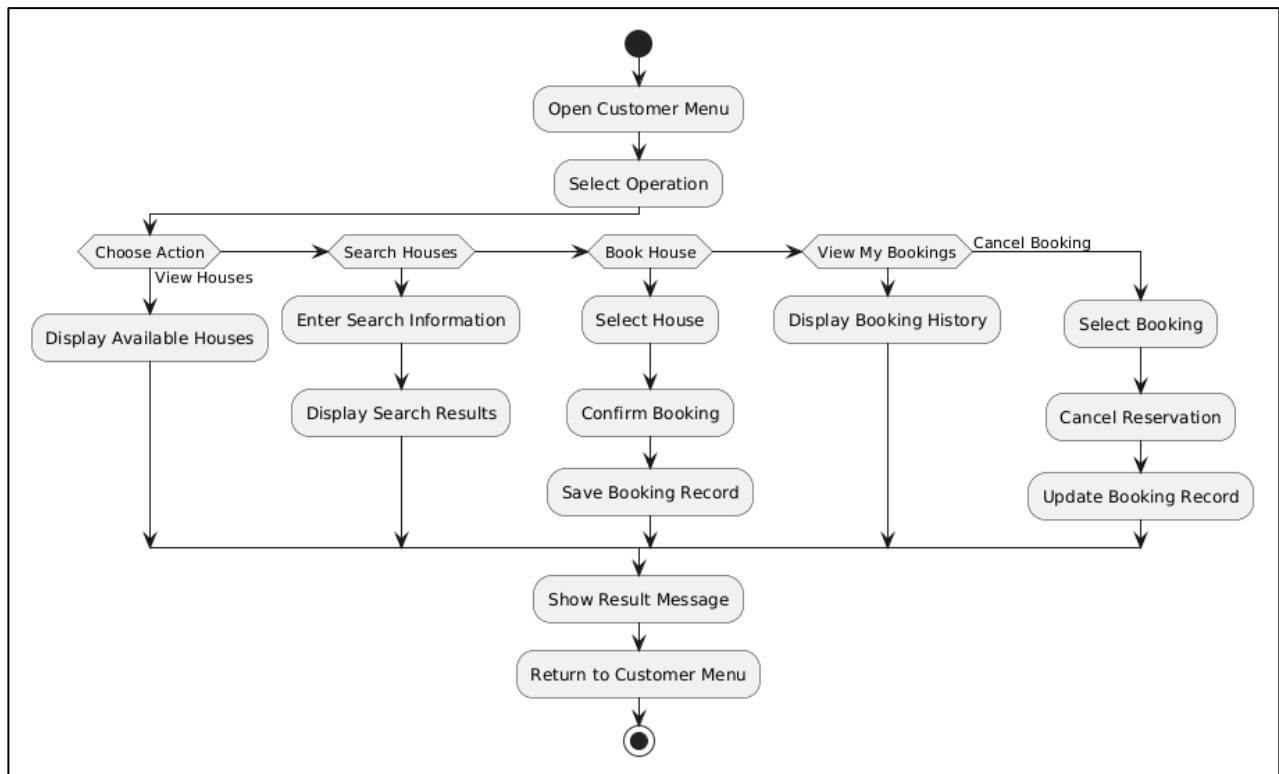
Full Activity Diagram of the System



Admin's Features Activity Diagram



Customer's Features Activity Diagram



11. System Workflow

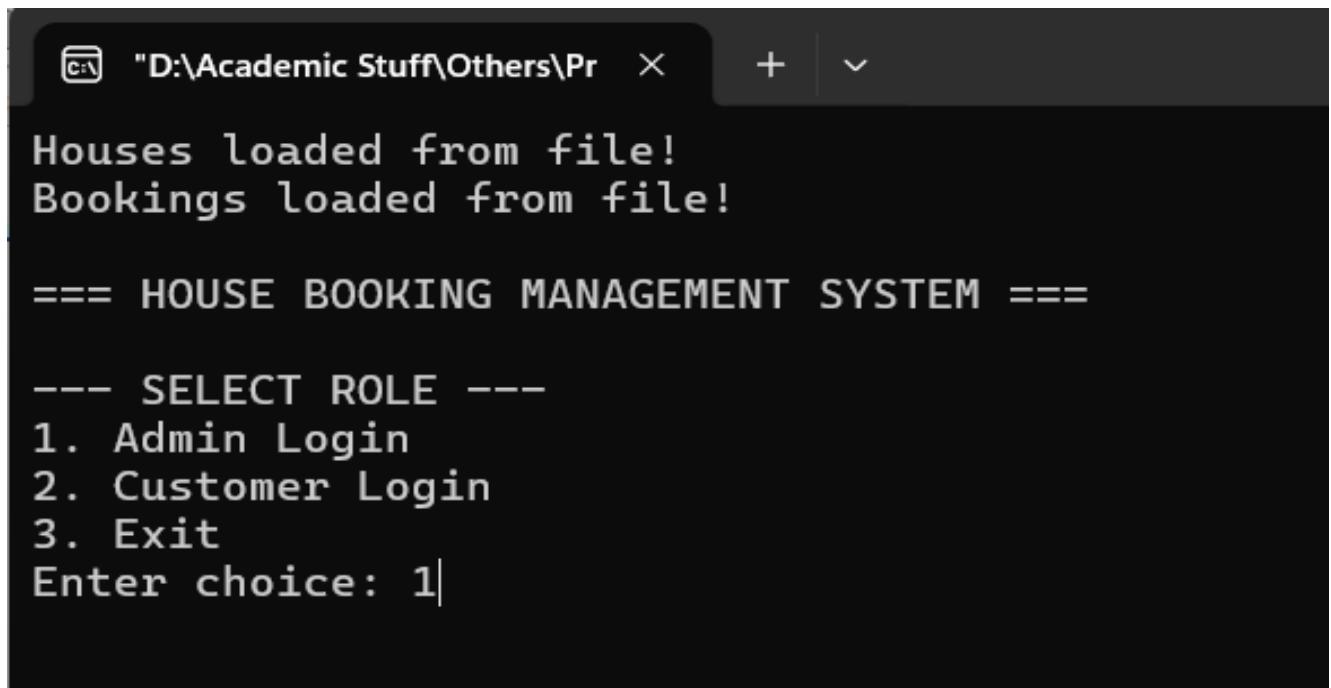
The workflow of the House Booking System begins when the user opens the application. The user then selects either the admin module or customer module.

In the admin module, administrators can manage house records by adding, updating, deleting, and viewing house information and bookings.

In the customer module, users can browse available houses, search for houses, make bookings, view booking history, and cancel reservations.

All booking and house information is stored using file handling, allowing the system to maintain records even after the program is closed.

12. Output



```
"D:\Academic Stuff\Others\Pr"
Houses loaded from file!
Bookings loaded from file!

=== HOUSE BOOKING MANAGEMENT SYSTEM ===

--- SELECT ROLE ---
1. Admin Login
2. Customer Login
3. Exit
Enter choice: 1|
```

Fig-1: Main Menu


```
--- SELECT ROLE ---
1. Admin Login
2. Customer Login
3. Exit
Enter choice: 3
Saving data...
Houses saved to file!
Bookings saved to file!
Thank you! Goodbye!

Process returned 0 (0x0)   execution time : 178.188 s
Press any key to continue.
|
```

Fig-2: Exit & File Saving

```
--- SELECT ROLE ---
1. Admin Login
2. Customer Login
3. Exit
Enter choice: 1
Admin Login Required
Enter Password: Admin
Login successful!

=== ADMIN PANEL ===

1. Add House
2. View All Houses
3. View All Bookings
4. Delete House
5. Modify House
6. Logout
Enter choice: 2|
```

Fig-3: Admin Panel Options

```
--- SELECT ROLE ---
1. Admin Login
2. Customer Login
3. Exit
Enter choice: 2

Enter your name: Abcd

=== CUSTOMER PANEL (Welcome Abcd) ===

    1. View Available Houses
    2. Search Houses
    3. Book House
    4. View My Bookings
    5. Cancel Booking
    6. Back to Main
Enter choice: 3
Enter house ID: 2
Enter your name: Abcd
Enter phone: 0123456789
Enter booking date (DD/MM/YYYY): 27-05-2024
Booking successful! Booking ID: 4
Bookings saved to file!
Houses saved to file!
```

Fig-4: Customer Panel Options

13. Limitations of the System

The current version of the House Booking System has some limitations:

- The system is console-based and does not provide a graphical interface.
- No database management system is used.
- Multi-user access is not supported.
- No authentication or password security system is implemented.
- The project is suitable mainly for small-scale usage.

14. Future Improvements

The following improvements can be added in future versions of the project:

- Development of a graphical user interface (GUI)
- Database integration for better data management
- User authentication and login system
- Online payment gateway integration
- Advanced search and filtering features
- Multi-user and network support
- Improved security and data validation

15. Conclusion

The House Booking System successfully demonstrates the implementation of a console-based rental and booking management application using the C programming language. The project simplifies house management and booking operations for both administrators and customers.

Through this project, important programming concepts such as file handling, structures, functions, and menu-driven application development were practically implemented. The system provides an organized approach for maintaining house and booking records efficiently.

This project also helped in developing problem-solving skills and gaining practical experience in software development.